# Mock 8: Wrong-Answer Notes

CFA Level II · 14 wrong answers and 6 right answers you were not fully sure of, one per page

| **30 / 44** | **23 / 36** | **7 / 8** |
|---|---|---|
| Mock 8 score (68.2%) | New vignettes (63.9%) | Re-done vignettes (87.5%) |

### Score by topic

| Topic | Score |
|---|---|
| Quantitative Methods | 0/4 |
| Derivatives | 1/4 |
| Economics | 2/4 |
| Financial Statement Analysis | 2/4 |
| Fixed Income | 3/4 |
| Corporate Issuers | 3/4 |
| Ethical and Professional Standards | 3/4 |
| Equity Valuation | 8/8 |
| Portfolio Management | 4/4 |
| Alternative Investments | 4/4 |

### Questions at a glance

| Q | Topic | You | Ans | |
|---|---|---|---|---|
| Q1 | Two-period binomial call: the price at purchase is c0, two steps back from the payoffs | C | B | wrong |
| Q2 | American put: arbitrage when the price is below the exercise value | B | A | wrong |
| Q4 | Implied volatility pattern across strikes: skew versus smile versus term structure | C | B | wrong |
| Q6 | Break-even inflation: nominal yield minus real yield of the SAME maturity | A | C | wrong |
| Q8 | Credit bonds in a recession turning to recovery: low-quality spreads narrow most; expected loss = PD × (1 − recovery) | C | B | wrong |
| Q10 | Credit spread from the tree: VND, CVA with LGD = 1 − recovery, fair value, YTM, spread | A | B | wrong |
| Q11 | What narrows a credit spread: less expected loss (lower PD or lower LGD) | C | C | right, not sure |
| Q13 | Log-linear trend: the current quarter is t = number of observations + 1 | A | C | wrong |
| Q14 | AR(1) mean-reverting level = b0 ÷ (1 − b1) | B | C | wrong |
| Q15 | Unit root: if the Dickey–Fuller test does not reject, difference the series | C | A | wrong |
| Q16 | AR(1) with b1 = 1: random walk with drift (b0 ≠ 0) | A | B | wrong |
| Q20 | H-model versus spreadsheet model | B | B | right, not sure |
| Q25 | Model choice: intangible assets and only occasional dividends point to FCFF | B | B | right, not sure |
| Q30 | Pension discount rate: higher than peers is aggressive | B | C | wrong |
| Q31 | Losses on asset sales every time: depreciation understated, so BOTH earnings quality and reporting quality are low | A | A | right, not sure |
| Q32 | Adjusting the pension discount rate down: obligation up, equity down | C | B | wrong |
| Q34 | ESG scoring: apply the rules item by item | C | A | wrong |

| | | | | |
|---|---|---|---|---|
| Q39 | Standard VI(C) Referral Fees: disclose, monitor the agents, notify clients at least quarterly | C | A | wrong |
| Q43 | Merger arbitrage gain in a stock-for-stock deal: convert the long shares, close the short | B | B | right, not sure |
| Q44 | Main risk of merger arbitrage: the deal fails | B | B | right, not sure |

### Patterns

- **Count the steps:** a two-year binomial price needs two discount steps (Q1); the forecast quarter after 24 observations is t = 25 (Q13); recovery of one-quarter means a loss given default of 75% (Q10).
- **Direction and sign:** a higher pension discount rate is aggressive (Q30) and, when adjusted, lowers book value per share (Q32); inflation uncertainty raises break-even inflation (Q6); unit root not rejected means difference the series (Q15).
- **Formula direction:** mean-reverting level = intercept ÷ (1 − slope) (Q14); expected loss uses (1 − recovery) (Q8); American put arbitrage compares the price with K − S (Q2).
- **Reading the pattern:** skew means one direction across strikes (Q4); random walk with drift has b1 = 1 and b0 ≠ 0 (Q16); “least likely” questions in Ethics ask for the one bad action (Q39).
- **Right answers you were not sure of** (Q11, 20, 25, 31, 43, 44): the one-line logic is kept on each page.

## Mock 8 Q1 Two-period binomial call: the price at purchase is c0, two steps back from the payoffs

You **C** · Answer **B**

### VIGNETTE (RELEVANT PART, UNEDITED)

Messer replies, "The binomial valuation model can be applied to the two-year European style index call options we purchased one year ago. The applicable underlying instrument is the German Blue Chip Equity price index, which excludes dividends. Exhibit 1 shows the option's characteristics at the time of purchase."

**Exhibit 1**

**Binomial Model Variables**

| | |
|---|---|
| u | 1.15 |
| d | 0.90 |
| π | 0.52 |
| Index price | EUR 720 |
| Strike price | EUR 750 |
| 1-year interest rate | 3% |
| S– | 648 |

**Question 1. Using the binomial valuation method and the data in Exhibit 1, the price Messer paid one year ago for the call option with a strike price of EUR750 is closest to:**

### OPTIONS

| | | | |
|---|---|---|---|
| **A** | EUR47.57. | | 0.48 × 102.08 ÷ 1.03 reverses the probabilities: π = 0.52 belongs to the UP branch. |
| **B** | EUR51.54. | ✓ Correct | Correct: c0 = 0.52 × 102.08 ÷ 1.03 = EUR51.54. |
| **C** | EUR102.08. | ✗ Your pick | Your pick: EUR102.08 is c+, the call value one year later (one step back). The price paid at purchase needs a second step back. |

**Trigger** “two-year European style index call options we purchased one year ago”
“Exhibit 1 shows the option's characteristics at the time of purchase”
**Rule next time: Price at purchase = value at time 0 of a 2-year option: payoffs at the end of year 2, then discount back TWICE (c+ is only the value after the first year).**

### FORMULA / RULE

> Terminal value: c++ = max(S++ − K, 0), c+− = c−− = 0
> c+ = [π × c++ + (1 − π) × c+−] ÷ (1 + r); c0 = [π × c+ + (1 − π) × c−] ÷ (1 + r)

### WHY THE ANSWER IS B

1. S++ = 720 × 1.15² = 952.2, so c++ = 952.2 − 750 = 202.2. The other two terminal payoffs are zero.
2. c+ = 0.52 × 202.2 ÷ 1.03 = 102.08; c− = 0 (both branches below are worthless).
3. c0 = 0.52 × 102.08 ÷ 1.03 = EUR51.54: **B**.

**Apply to other questions:** Count the periods first: a two-year option needs two discounting steps. The number you get after one step is a trap answer.

## Mock 8 Q2 American put: arbitrage when the price is below the exercise value

You **B** · Answer **A**

### VIGNETTE (RELEVANT PART, UNEDITED)

Messer explains, "Of course, the index has moved down 10% in the last 12 months."

Messer continues, "Twelve months ago, I noted that two-year American-style puts with a strike price of EUR750 cost EUR38.48. Using the information in Exhibit 1 and today's index value, the binomial valuation model calculates the current price of the put as EUR80.15. It is actually trading now above that price at EUR92."

**Question 2. Does the put option with a strike price of EUR750 currently offer an arbitrage opportunity?**

### OPTIONS

| | | | |
|---|---|---|---|
| **A** | Yes | ✓ Correct | Correct: the index is now 720 × 0.90 = 648, so exercise gives 750 − 648 = EUR102 against a price of EUR92: arbitrage. |
| **B** | No, because the market has bid up the price of the put | ✗ Your pick | Your pick: the price of the put is EUR92, still BELOW its exercise value EUR102, so the market has not bid it up enough. The binomial value of EUR80.15 is irrelevant to the arbitrage test. |
| **C** | No, because the put is deep in the money | | Being deep in the money does not remove the arbitrage; the test is price versus exercise value. |

**Trigger** “the index has moved down 10% in the last 12 months”
“two-year American-style puts with a strike price of EUR750”
“trading now above that price at EUR92”

**Rule next time: American put: compare its market price with the immediate exercise value K − S. If price < K − S, buy it, exercise and lock in the difference.**

### FORMULA / RULE

> American put lower bound: $p \geq \max(0, K - S)$
> Price < K − S: buy the put, buy the index, exercise: profit = (K − S) − price

### WHY THE ANSWER IS A

1. Index now = 648, strike 750, so exercise value = 102.
2. Market price = 92 < 102.
3. Buy the put (92) and exercise (receive 102): a risk-free gain of 10. Yes: **A**.

**Apply to other questions:** For American options always check the early-exercise value, not only the model value.

## Mock 8 Q4 Implied volatility pattern across strikes: skew versus smile versus term structure

You **C** · Answer **B**

VIGNETTE (RELEVANT PART, UNEDITED)

Messer concludes, "We also use the BSM model to calculate the implied volatility. The implied volatilities of the index options expiring in one year are shown in Exhibit 2."

**Exhibit 2**

**Implied Volatility Curve**

| Strike Price | Implied Volatility |
|---|---|
| 700 | 18.7 |
| 710 | 17.98 |
| 720 | 17.38 |
| 730 | 16.69 |
| 740 | 15.83 |
| 750 | 15.40 |
| 760 | 14.50 |
| 770 | 14.03 |
| 780 | 13.21 |
| 790 | 12.11 |
| 800 | 11.09 |

**Question 4. Exhibit 2 shows that the implied volatility of the EUR700 option (18.7) is higher than that of the EUR800 option (11.09), and falls steadily as the strike price rises. This pattern is best described as:**

OPTIONS

| | | | |
|---|---|---|---|
| **A** | a volatility smile, in which implied volatility is higher at both low and high strike prices. | | Correct: implied volatility falls as the strike rises: a volatility skew. |
| **B** | a volatility skew, in which implied volatility is higher for lower strike prices. | ✓ Correct | A smile is U-shaped (high at both ends); Exhibit 2 only falls. |
| **C** | the term structure of volatility, in which implied volatility differs by time to expiration. | ✗ Your pick | Your pick: term structure of volatility compares different EXPIRATIONS. All options in Exhibit 2 expire in one year; only the strike changes. |

**Trigger** "falls steadily as the strike price rises"
"index options expiring in one year"
**Rule next time: Skew = implied volatility moves one way as the strike rises (higher for low strikes). Smile = U-shape. Term structure = different expiries.**

FORMULA / RULE

Volatility skew: implied volatility vs strike, one-directional
Volatility smile: U-shaped across strikes
Term structure of volatility: implied volatility vs time to expiration

WHY THE ANSWER IS B

1. Same expiry (one year), different strike prices: strike is the variable.
2. 18.7 at 700 falling to 11.09 at 800: one direction only.
3. Volatility skew: **B**.

**Apply to other questions:** Ask: what changes across the table, the strike or the maturity? Strike = skew or smile; maturity = term structure.

## Mock 8 Q6 Break-even inflation: nominal yield minus real yield of the SAME maturity

You **A** · Answer **C**

### VIGNETTE (RELEVANT PART, UNEDITED)

At the investment committee meeting, Simmons recommends purchasing United Kingdom inflation-linked government bonds. In her discussion, she makes the following statements about inflation-linked bonds:

Statement 1: “Increases in uncertainty about future inflation are associated with lower break-even inflation rates.”

Statement 2: “The break-even inflation rate incorporates the yield difference between longer dated and shorter dated inflation-linked bonds.”

Statement 3: “The difference between nominal and inflation-linked bond yields reflects the expectations of and uncertainty about the quantity of goods and services investors will be able to consume in the future.”

**Question 6. Which of Simmons’ statements concerning inflation-linked bonds is most likely correct?**

### OPTIONS

| | | | |
|---|---|---|---|
| **A** | Statement 1 | ✗ Your pick | Your pick: Statement 1 is reversed: more uncertainty about inflation raises the break-even rate. |
| **B** | Statement 2 | | Statement 2 describes a term spread between two inflation-linked bonds, not the break-even rate (nominal minus inflation-linked of the same maturity). |
| **C** | Statement 3 | ✓ Correct | Correct: the yield gap reflects inflation expectations and the uncertainty about the goods and services investors can consume. |

**Trigger** “reflects the expectations of and uncertainty about the quantity of goods and services”
“Increases in uncertainty about future inflation are associated with lower break-even inflation rates.”

**Rule next time: Higher inflation uncertainty = higher break-even inflation (inflation risk premium). Break-even = nominal yield − real yield, same maturity.**

### FORMULA / RULE

> Break-even inflation rate = yield on nominal bond − yield on inflation-linked bond (same maturity)
> BEI = expected inflation + inflation risk premium

### WHY THE ANSWER IS C

1. Statement 1: uncertainty ↑ → risk premium ↑ → BEI ↑ (so “lower” is wrong).
2. Statement 2: the BEI compares nominal and linked bonds of the same maturity, not long versus short linked bonds.
3. Statement 3 is correct: **C**.

**Apply to other questions:** Read the direction word in each statement (higher or lower) before accepting it.

## Mock 8 Q8 Credit bonds in a recession turning to recovery: low-quality spreads narrow most; expected loss = PD × (1 − recovery)

You **C** · Answer **B**

VIGNETTE (RELEVANT PART, UNEDITED)

Lloyd Redfield, a portfolio manager on the foreign bond team, asks Simmons to review the current business cycle and yield curve in Australia. The Australian economy entered a recession approximately 6 months ago, and the Australian Central Bank has been reducing its policy rate. Simmons expects the Australian economy to gradually improve over the next 6–12 months.

Given the ongoing recession in Australia, Redfield is reviewing Australian corporate bonds with the same five-year maturity held in the foreign bond strategy. The three bonds being reviewed are listed in Exhibit 2. Redfield agrees with Simmons' assessment of the Australian economy.

**Exhibit 2 Australia Corporate Bond Holdings**

| Bond | Credit Rating | Sector | Probability of Default | Estimated Recovery Rate | Spread over Government Bonds |
|---|---|---|---|---|---|
| A | A | Consumer Staples | 0.5% | 40% | 200 bps |
| B | BBB– | Consumer Discretionary | 2.0% | 35% | 350 bps |
| C | BB | Industrial | 3.5% | 45% | 650 bps |

Based on this information, Redfield makes the following comments to Simmons regarding the Australian bonds:

Comment 1: Relative to bond B and bond C, bond A has likely experienced the worst price return over the last 9 months.

Comment 2: Based on your economic forecasts, in 6–12 months overall credit spreads should narrow and the rate of improvement will likely be greatest for bond C.

Comment 3: You can calculate the expected loss by multiplying the probability of default by the recovery rate.

**Question 8. Which of Redfield's comments regarding the Australian corporate bonds in Exhibit 2 is most likely correct?**

OPTIONS

| | | | |
|---|---|---|---|
| **A** | Comment 1 | | Comment 1 is reversed: in a downturn the good credit (bond A) outperforms; spreads between low and high quality widen. |
| **B** | Comment 2 | ✓ Correct | Correct: as the economy improves spreads narrow, and bond C (lowest rating, widest spread) improves most. |
| **C** | Comment 3 | ✗ Your pick | Your pick: Comment 3 multiplies PD by the recovery rate. Expected loss uses LOSS given default (1 − recovery): bond B = 2% × 65% = 1.3%. |

**Trigger** "Simmons expects the Australian economy to gradually improve over the next 6–12 months"
"overall credit spreads should narrow"
"multiplying the probability of default by the recovery rate"

**Rule next time: Spreads narrowing: the weakest credits (widest spread, highest PD) improve the most. Expected loss = PD × LGD = PD × (1 − RR), never PD × RR.**

FORMULA / RULE

> Expected loss = probability of default × loss given default; LGD = 1 − recovery rate
> Recession: spreads widen, more for low quality. Recovery: spreads narrow, more for low quality

WHY THE ANSWER IS B

1. Comment 1: wrong, A is the highest quality and does best in a recession.
2. Comment 2: right, weaker credits gain most when spreads narrow.
3. Comment 3: wrong formula (needs 1 − RR). Answer **B**.

**Apply to other questions:** "Recovery rate" in a formula usually appears as (1 − recovery rate). Check which one the formula needs.

## Mock 8 Q10 Credit spread from the tree: VND, CVA with LGD = 1 − recovery, fair value, YTM, spread

You **A** · Answer **B**

### VIGNETTE (RELEVANT PART, UNEDITED)

Marangoni next asks Brandt to analyze a three-year bond, issued by Norvalle S.A., using an arbitrage-free framework. The bond's coupon rate is 4%, with interest paid annually and a par value of 100. In the analysis, Brandt makes the following three assumptions:

The annual interest rate volatility is 15%.

The recovery rate is one-quarter of the exposure each period.

The annual probability of default each year is 1.50%.

Selected information on benchmark government bonds for the Norvalle bond is presented in Exhibit 2, and the relevant binomial interest rate tree is presented in Exhibit 3.

**Exhibit 2**

Par Curve Rates for Annual Payment Benchmark Government Bonds

| Maturity | Coupon rate | Price | Discount factor | Spot rate | Forward rate |
|---|---|---|---|---|---|
| 1 | 2.50% | 100 | 0.975610 | 2.5000% | 2.5000% |
| 2 | 3.30% | 100 | 0.936888 | 3.3133% | 4.1331% |
| 3 | 4.00% | 100 | 0.887981 | 4.0396% | 5.5076% |

**Exhibit 3**

One-Year Binomial Interest Rate Tree for 15% Volatility (risk-neutral probability of 50% on each branch)

| Date | 0 | 1 (top) | 1 (bottom) | 2 (top) | 2 (middle) | 2 (bottom) |
|---|---|---|---|---|---|---|
| Rate | 2.5000% | 4.7526% | 3.5208% | 7.2932% | 5.4029% | 4.0026% |

Brandt estimates the value of the Norvalle bond assuming no default (VND) as well as the fair value of the bond. Brandt then estimates the bond's yield to maturity and the bond's credit spread over the benchmark in Exhibit 2. Brandt asks Marangoni, "What might cause the bond's credit spread to decrease?"

**Question 10. Based on Brandt's assumptions and Exhibits 2 and 3, the credit spread on the Norvalle bond is closest to:**

### OPTIONS

| | | | |
|---|---|---|---|
| **A** | 0.7796%. | ✗ Your pick | Your pick: 0.7796% results from LGD = 50% instead of 75% (recovery is one-quarter, not one-half). |
| **B** | 1.1782%. | ✓ Correct | Correct: 1.1782%. |
| **C** | 1.5828%. | | 1.5828% assumes no recovery (LGD = 100%). |

**Trigger** “The recovery rate is one-quarter of the exposure each period.”
“The annual probability of default each year is 1.50%.”
“the relevant binomial interest rate tree is presented in Exhibit 3”

**Rule next time: Recovery of one-quarter means LGD = 75% of the exposure. Then CVA = Σ DF × PD × LGD × exposure; fair value = VND − CVA; the spread = YTM at fair value − benchmark yield.**

### FORMULA / RULE

Cumulative PD_t = 1 − (1 − PD)^t; marginal PD_t = (1 − PD)^(t−1) × PD (survival to t−1 and default in t)
CVA = Σ DF_t × marginal PD_t × LGD_t × expected exposure_t; fair value = VND − CVA
Spread = YTM(fair value) − benchmark YTM

### WHY THE ANSWER IS B

1. VND from the tree = 100.0000 (coupon 4%, discount at each tree rate).
2. Expected exposures 102.5000, 102.5665, 104.0000; LGD = 0.75 × exposure; CVA = 1.1250 + 1.0648 + 1.0080 = 3.1978.
3. Fair value 96.8022 → YTM 5.1782%; spread = 5.1782% − 4.0000% = 1.1782%: **B**.

**Apply to other questions:** Translate the recovery sentence into LGD before doing anything else. Recovery 25% → LGD 75%.

## Mock 8 Q11 What narrows a credit spread: less expected loss (lower PD or lower LGD)

You **C** · Answer **C**

### VIGNETTE (RELEVANT PART, UNEDITED)

Brandt estimates the value of the Norvalle bond assuming no default (VND) as well as the fair value of the bond. Brandt then estimates the bond's yield to maturity and the bond's credit spread over the benchmark in Exhibit 2. Brandt asks Marangoni, "What might cause the bond's credit spread to decrease?"

**Question 11. The most appropriate response to Brandt's question relating to the credit spread is:**

### OPTIONS

| | | | |
|---|---|---|---|
| **A** | an increase in the loss given default. | | A higher LGD raises the CVA and widens the spread. |
| **B** | an increase in the probability of default. | | A higher probability of default raises the CVA and widens the spread. |
| **C** | a decrease in the risk-neutral probability of default. | ✓ Correct | Correct: a lower risk-neutral probability of default lowers the CVA, raises the fair value and lowers the spread. |

**Trigger** “What might cause the bond's credit spread to decrease?”

**Rule next time: Spread falls when the CVA falls: lower probability of default or lower loss given default.**

### FORMULA / RULE

> Fair value = VND − CVA; larger CVA → lower fair value → higher YTM → wider spread

### WHY THE ANSWER IS C

1. Spread ↓ needs fair value ↑ (towards VND).
2. That needs CVA ↓: lower PD or lower LGD.
3. Lower risk-neutral PD: **C**.

**Apply to other questions:** Right in this mock; keep the logic: CVA up = spread up.

## Mock 8 Q13 Log-linear trend: the current quarter is t = number of observations + 1

You **A** · Answer **C**

### VIGNETTE (RELEVANT PART, UNEDITED)

First, Taylor examines the quarterly sales (in $ millions) of the company for the past six years (24 observations) to forecast sales for the current quarter. Taylor uses Model A, a log-linear model (ln yt = b0 + b1t + εt) where yt represents quarterly sales for quarter t. The regression estimates for the intercept and the trend coefficient are 5.74820 and 0.03795, respectively.

**Question 13. Based on Model A, the predicted value of Maclaurin's sales for the current quarter is closest to:**

### OPTIONS

| | | | |
|---|---|---|---|
| **A** | $326 million. | ✗ Your pick | Your pick: $326 million = exp(5.74820 + 0.03795 × 1); that is t = 1, the start of the sample, not the current quarter. |
| **B** | $780 million. | | t = 24 would be the last quarter of the sample, exp(...) = 780. |
| **C** | $810 million. | ✓ Correct | Correct: t = 25: exp(5.74820 + 0.03795 × 25) = $810 million. |

**Trigger** “for the past six years (24 observations) to forecast sales for the current quarter”
“a log-linear model (ln yt = b0 + b1t + εt)”

**Rule next time: 24 observations in the sample, so the quarter being forecast is t = 25. In a log-linear model take the exponential at the end.**

### FORMULA / RULE

ln y_t = b0 + b1 t, so y_t = exp(b0 + b1 t)
t for the first out-of-sample period = n + 1

### WHY THE ANSWER IS C

1. Six years × 4 quarters = 24 observations, so t = 25.
2. ln y = 5.74820 + 0.03795 × 25 = 6.69695.
3. y = exp(6.69695) = 809.93: **C**.

**Apply to other questions:** Write t down first, then compute. And exponentiate in a log-linear model.

## Mock 8 Q14 AR(1) mean-reverting level = b0 ÷ (1 − b1)

You **B** · Answer **C**

### VIGNETTE (RELEVANT PART, UNEDITED)

Maclaurin's earnings are substantially export-driven. To understand the impact of foreign exchange rates on earnings, Taylor builds Model B, an autoregressive AR(1) model, for the exchange rate between Maclaurin's domestic currency and that of its primary export market. The regression estimates for the intercept and the coefficient of the first lag are 0.3000 and 0.9093, respectively.

**Question 14. For Model B, the mean-reverting level is closest to:**

### OPTIONS

| | | | |
|---|---|---|---|
| **A** | 1.30. | | 1.30 = 0.9093 ÷ 0.7000 swaps b0 and b1. |
| **B** | 3.03. | ✗ Your pick | Your pick: 3.03 = 0.9093 ÷ 0.3000 is slope ÷ intercept, which is not the formula. |
| **C** | 3.31. | ✓ Correct | Correct: 0.3000 ÷ (1 − 0.9093) = 3.31. |

**Trigger** “Model B, an autoregressive AR(1) model”
“0.3000 and 0.9093”

**Rule next time: Mean-reverting level = intercept ÷ (1 − slope coefficient).**

### FORMULA / RULE

> x_t = b0 + b1 x_{t−1}; mean-reverting level = b0 ÷ (1 − b1) (requires |b1| < 1)

### WHY THE ANSWER IS C

1. 1 − 0.9093 = 0.0907.
2. 0.3000 ÷ 0.0907 = 3.31: **C**.

**Apply to other questions:** Intercept on top, (1 − slope) on the bottom.

## Mock 8 Q15 Unit root: if the Dickey–Fuller test does not reject, difference the series

You **C** · Answer **A**

### VIGNETTE (RELEVANT PART, UNEDITED)

Taylor builds another AR(1) model (Model C) for the exchange rate between Maclaurin's domestic currency and that of its secondary export market. Taylor suspects that the time series used for Model C may have a unit root and he runs a Dickey–Fuller test.

**Question 15. Taylor suspects that the exchange rate series used for Model C may have a unit root. If his Dickey–Fuller test does not reject its null hypothesis, he should most likely:**

### OPTIONS

| | | | |
|---|---|---|---|
| **A** | first-difference the series before estimating an autoregressive model. | ✓ Correct | Correct: first-difference the series. |
| **B** | add a second lag to Model C, which removes the unit root. | | Adding lags does not remove a unit root. |
| **C** | estimate the AR(1) model on the original series, because a unit root makes the estimates more precise. | ✗ Your pick | Your pick: estimating the AR(1) on the original series is the mistake. With a unit root the estimates are unreliable (not more precise). |

**Trigger** “may have a unit root”
“does not reject its null hypothesis”
**Rule next time: Dickey–Fuller null = unit root. Not rejecting means the series is not covariance stationary: first-difference it before using AR.**

### FORMULA / RULE

> H0: unit root (b1 = 1). Fail to reject → not covariance stationary → transform (first difference)

### WHY THE ANSWER IS A

1. Null = unit root; not rejected → the unit root is still there.
2. A series with a unit root has no stable mean, so AR estimates are not valid.
3. First-difference: **A**.

**Apply to other questions:** Dickey–Fuller: a LOW p-value (reject) is the good news.

## Mock 8 Q16 AR(1) with b1 = 1: random walk with drift (b0 ≠ 0)

You **A** · Answer **B**

### VIGNETTE (RELEVANT PART, UNEDITED)

Before making his recommendation on the stock, Taylor also wants to model the movement of Maclaurin's share price (in natural logs). He uses an AR(1) model (Model D). After performing the relevant necessary transformations and reviewing the regression statistics, Taylor concludes that the intercept is 0.25 and the coefficient of the first lag is 1.

**Question 16. Taylor's regression estimates for Model D indicate that the time series of Maclaurin's share price (in natural logs) is most likely:**

### OPTIONS

| | | | |
|---|---|---|---|
| **A** | a covariance stationary series. | ✗ Your pick | Your pick: with b1 = 1 the mean-reverting level b0 ÷ (1 − b1) is undefined, so the series cannot be covariance stationary. |
| **B** | a random walk with drift. | ✓ Correct | Correct: b1 = 1 and b0 = 0.25 ≠ 0: random walk with drift. |
| **C** | a simple random walk. | | A simple random walk would have b0 = 0. |

**Trigger** "the intercept is 0.25 and the coefficient of the first lag is 1"

**Rule next time: b1 = 1 and b0 ≠ 0 → random walk with drift; b1 = 1 and b0 = 0 → simple random walk. Either has a unit root and is NOT covariance stationary.**

### FORMULA / RULE

Covariance stationary needs |b1| < 1

Random walk: b0 = 0, b1 = 1; with drift: b0 ≠ 0, b1 = 1

### WHY THE ANSWER IS B

1. b1 = 1 → unit root → not covariance stationary.
2. b0 = 0.25 ≠ 0 → drift.
3. Random walk with drift: **B**.

**Apply to other questions:** Check b1 first (stationary or not), then b0 (drift or not).

## Mock 8 Q20 H-model versus spreadsheet model

You **B** · Answer **B**

VIGNETTE (RELEVANT PART, UNEDITED)

Jackson has been researching how to value shares using a dividend growth model if the transition from one growth rate to a lower one is gradual. In that scenario, he understands he would use the H-model to evaluate the shares. If, however, the difference in growth rates is large, or the length of the high growth rate is extraordinarily long, then he would have to use a spreadsheet model instead.

**Question 20. Jackson's understanding of when he would switch from an H-model to a spreadsheet model is best described as:**

OPTIONS

| | | | |
|---|---|---|---|
| **A** | incorrect with respect to the difference in growth rates. | | Incorrect: his view of the growth-rate difference is right. |
| **B** | correct. | ✓ Correct | Correct: both conditions he names are right. |
| **C** | incorrect with respect to the length of the first growth period. | | Incorrect: his view of the length of the high-growth period is right. |

**Trigger** "if the transition from one growth rate to a lower one is gradual"
"the difference in growth rates is large, or the length of the high growth rate is extraordinarily long"

**Rule next time: H-model: linear decline from a high to a lower growth rate. A very large growth difference or a very long high-growth period needs a spreadsheet model.**

FORMULA / RULE

> H-model: V0 = D0(1 + gL)/(r − gL) + D0 × H × (gS − gL)/(r − gL), H = half the length of the transition

WHY THE ANSWER IS B

1. The H-model approximates a linear transition.
2. It becomes inaccurate for large growth gaps or very long extraordinary growth.
3. Jackson is correct: **B**.

**Apply to other questions:** Right in this mock; keep the two limits: big growth difference, long high-growth period.

## Mock 8 Q25 Model choice: intangible assets and only occasional dividends point to FCFF

You **B** · Answer **B**

### VIGNETTE (RELEVANT PART, UNEDITED)

Company A is an employment services firm with no debt and has fixed assets consisting primarily of computers, servers, and commercially available software. Many of the assets are intangible, including human capital. The company has a history of occasionally paying a special cash dividend.

**Question 25. Based on Company A's characteristics, which of the following absolute valuation models is most appropriate for valuing that company?**

### OPTIONS

| | | | |
|---|---|---|---|
| **A** | Asset based | | Asset-based works for companies with mostly tangible, marketable assets. |
| **B** | Free cash flow to the firm | ✓ Correct | Correct: FCFF. |
| **C** | Dividend discount | | Dividend discount needs a consistent, predictable dividend; special dividends are occasional. |

**Trigger** "no debt"
"Many of the assets are intangible"
"occasionally paying a special cash dividend"

**Rule next time: Intangible assets rule out an asset-based model; irregular special dividends rule out the dividend model; use free cash flow (FCFF/FCFE).**

### FORMULA / RULE

Asset-based: tangible, marketable assets
DDM: consistent dividend policy
FCFF/FCFE: no regular dividends or assets mostly intangible

### WHY THE ANSWER IS B

1. Assets are intangible → no asset-based.
2. Dividends only occasional → no DDM.
3. FCFF: **B**.

**Apply to other questions:** Right in this mock; keep the clue words: intangible, occasional special dividend.

## Mock 8 Q30 Pension discount rate: higher than peers is aggressive

You **B** · Answer **C**

### VIGNETTE (RELEVANT PART, UNEDITED)

Thorsell next examines Coleridge's underfunded defined benefit pension plan and observes that the plan's discount rate assumption is 0.5% higher than its industry competitors' discount rates. For comparison purposes, Thorsell adjusts Coleridge's financial statements before computing the company's financial ratios.

**Question 30. Compared with its industry competitors, Coleridge's pension discount rate assumption is most likely:**

### OPTIONS

| | | | |
|---|---|---|---|
| **A** | neutral, because the plan is underfunded in any case. | | The size of the deficit depends on the assumption, so “neutral” is wrong. |
| **B** | conservative, because it raises the reported pension expense. | ✗ Your pick | Your pick: a higher discount rate LOWERS, not raises, the obligation and expense, so it is not conservative. |
| **C** | aggressive, because it lowers the reported pension obligation. | ✓ Correct | Correct: aggressive, because it lowers the reported obligation. |

**Trigger** “the plan's discount rate assumption is 0.5% higher than its industry competitors' discount rates”

**Rule next time: A higher discount rate lowers the present value of the pension obligation (and the pension expense). Higher than peers = aggressive.**

### FORMULA / RULE

> Higher discount rate → lower PBO and lower pension cost (aggressive)
> Lower discount rate → higher PBO and higher cost (conservative)

### WHY THE ANSWER IS C

1. Discounting future benefits at a higher rate shrinks the present value.
2. Smaller obligation and expense flatter the company.
3. Aggressive: **C**.

**Apply to other questions:** Link: discount rate and PV move in opposite directions.

## Mock 8 Q31 Losses on asset sales every time: depreciation understated, so BOTH earnings quality and reporting quality are low

You **A** · Answer **A**

### VIGNETTE (RELEVANT PART, UNEDITED)

> Next, Thorsell notes that, consistent with past years, Coleridge has generated only accounting losses when it sells any one of its long-term assets. The losses are presented on the income statement after operating income, and when they occur they are significant in amount.

**Question 31. With respect to the sale of its long-term assets, Coleridge most likely displays:**

### OPTIONS

| | | | |
|---|---|---|---|
| **A** | both low earnings quality and low financial reporting quality. | ✓ Correct | Correct: both are low. |
| **B** | low earnings quality only. | | Earnings quality is low, but so is the financial reporting quality: B is incomplete. |
| **C** | low financial reporting quality only. | | Reporting quality is low, but so is earnings quality: C is incomplete. |

**Trigger** “has generated only accounting losses when it sells any one of its long-term assets”

**Rule next time: Systematic losses on disposals mean depreciation was too low in earlier years: income was overstated (low earnings quality) and the numbers did not faithfully represent the economics (low reporting quality).**

### FORMULA / RULE

> Low reporting quality: not faithful or biased accounting choices
> Low earnings quality: earnings not sustainable or overstated

### WHY THE ANSWER IS A

1. Only losses on every sale: depreciation lives too long or expense too low.
2. Operating income overstated each year → low earnings quality.
3. Biased estimate → low reporting quality too: **A**.

**Apply to other questions:** Right in this mock; keep the pattern: consistent losses on disposal = understated depreciation.

## Mock 8 Q32 Adjusting the pension discount rate down: obligation up, equity down

You **C** · Answer **B**

### VIGNETTE (RELEVANT PART, UNEDITED)

Thorsell next examines Coleridge's underfunded defined benefit pension plan and observes that the plan's discount rate assumption is 0.5% higher than its industry competitors' discount rates. For comparison purposes, Thorsell adjusts Coleridge's financial statements before computing the company's financial ratios.

**Question 32. With respect to the defined benefit pension plan assumption, which of the following will decrease after Thorsell's adjustment?**

### OPTIONS

| | | | |
|---|---|---|---|
| **A** | Financial leverage ratio | | Financial leverage (assets ÷ equity) increases. |
| **B** | Book value per share | ✓ Correct | Correct: book value per share decreases (equity down, shares unchanged). |
| **C** | Debt-to-equity ratio | ✗ Your pick | Your pick: debt-to-equity increases (debt up, equity down); it does not decrease. |

**Trigger** “Thorsell adjusts Coleridge's financial statements before computing the company's financial ratios” “0.5% higher than its industry competitors'”

**Rule next time: Her adjustment lowers the discount rate to the peer level: the obligation rises, equity falls through AOCI, debt rises. Ratios with equity in the denominator rise; per-share equity falls.**

### FORMULA / RULE

Lower discount rate → higher PBO → bigger deficit → lower equity, higher debt

Equity ↓: BVPS ↓, D/E ↑, leverage ↑

### WHY THE ANSWER IS B

1. Peers use a lower discount rate, so the adjustment lowers Coleridge's rate by 0.5%.
2. Obligation ↑, equity ↓, shares unchanged.
3. BVPS decreases: **B**.

**Apply to other questions:** Sketch the direction of debt and equity first, then each ratio.

## Mock 8 Q34 ESG scoring: apply the rules item by item

You **C** · Answer **A**

VIGNETTE (RELEVANT PART, UNEDITED)

**Exhibit 1**

| | ChemiKa | PlastiKa |
|---|---|---|
| Ownership | Concentrated | Dispersed |
| Voting power | Concentrated | Dispersed |
| Total number of directors | 19 | 8 |
| Number of independent directors | 12 | 4 |
| Two-tiered board | No | Yes |
| CEO duality | Yes | No |
| Average board tenure | 8 years | 14 years |
| **Independent committees:** | | |
| Audit | Yes | No |
| Compensation | Yes | Yes |
| Director selection | No | Yes |

To evaluate the corporate governance policies and procedures of the two companies, Tyler considers the rankings provided by an independent ESG data analysis firm specializing in board and governance structure. The contributions of each board member are only considered after at least one year of service.

The ESG data analysis firm uses a scoring system awarding one point for favorable characteristics among the following:

CEO duality is absent.

Average board tenure is below ten years.

For board independence, a two-tiered board is considered favorable. A simple majority of independent directors is generally favorable. However, in a one-tiered board which is considered unfavorable, a 60% majority of independent directors is the favorable threshold.

Independent committees for audit, compensation and director selection are present (1 point for the presence of each independent committee).

**Question 34. Based on the ESG data analysis firm's scoring system and Exhibit 1, compared to PlastiKa, ChemiKa most likely ranks:**

OPTIONS

| | | | |
|---|---|---|---|
| **A** | the same. | ✓ Correct | Correct: both score the same. |
| **B** | lower. | | Lower would need ChemiKa to score less than PlastiKa; both reach 4 points. |
| **C** | higher. | ✗ Your pick | Your pick: higher would need ChemiKa to score more than PlastiKa; both reach 4 points. |

**Trigger** “The ESG data analysis firm uses a scoring system awarding one point for favorable characteristics”
“in a one-tiered board which is considered unfavorable, a 60% majority of independent directors is the favorable threshold”

**Rule next time: Score each listed feature separately: board independence (two-tiered board, or 60% independents in a one-tiered board), no CEO duality, tenure under 10 years, and each independent committee. Ownership and voting power are not scored.**

FORMULA / RULE

> Board independence: two-tiered board = favorable; one-tiered board needs 60% independents
> 1 point each: no CEO duality, tenure under 10 years, audit, compensation, director selection committees

WHY THE ANSWER IS A

1. ChemiKa: one-tiered board with 12 ÷ 19 = 63.2% independents (≥ 60%) = 1; CEO duality 0; tenure 8 years 1; audit 1, compensation 1, selection 0: total 4.
2. PlastiKa: two-tiered board 1; no CEO duality 1; tenure 14 years 0; audit 0, compensation 1, selection 1: total 4.
3. Same ranking: **A**.

**Apply to other questions:** Build a two-column score table before choosing; do not stop at the first favorable feature.

## Mock 8 Q39 Standard VI(C) Referral Fees: disclose, monitor the agents, notify clients at least quarterly

You **C** · Answer **A**

### VIGNETTE (RELEVANT PART, UNEDITED)

> Second, let me explain how we select investments. I'll give you the prospectus, too, for your further reference, because it contains all the details, and they're quite extensive. We identify potential investments from around the globe through our own analysts' research within the education and health care industries. We also use third-party agents to find potential investments. If we make an investment in the companies they present to us, we pay them a finder's fee. One such agent we use is Make-a-Difference Consultants, an asset consulting firm that has adopted the CFA Institute Code and Standards.

**Question 39. Given O'Connor's second statement to Jensen, which further action should O'Connor least likely take to comply with required or recommended procedures for the CFA Institute Standards of Professional Conduct?**

### OPTIONS

| | | | |
|---|---|---|---|
| **A** | Notify clients of the finder's fee policy on a semiannual basis. | ✓ Correct | Correct (the action least likely to be taken): notify clients SEMIANNUALLY is too infrequent; at least quarterly is recommended. |
| **B** | Monitor the consultants' compliance policy execution. | | Monitoring the consultants is expected (Standard I(A): do not assist others in a violation). |
| **C** | Disclose the finder's fee arrangement in the fund prospectus. | ✗ Your pick | Your pick: disclosing the finder's fee arrangement in the prospectus is exactly what the Standard requires. |

**Trigger** "If we make an investment in the companies they present to us, we pay them a finder's fee."
"least likely"

**Rule next time: "Least likely" means pick the action that is NOT required or recommended. Recommended procedures for referral fees: disclose them to clients and prospects, and notify clients on a regular basis (at least quarterly).**

### FORMULA / RULE

> Standard VI(C): disclose to the employer, clients and prospective clients any compensation or benefit paid for referrals, before the service is provided
> Recommended: notify clients at least quarterly of the finder's fee policy

### WHY THE ANSWER IS A

1. The question asks for the LEAST likely compliant step.
2. Disclosure in the prospectus and monitoring the agents are both required or recommended.
3. Semiannual notification is weaker than the recommended quarterly: **A**.

**Apply to other questions:** Flip the logic in "least likely" questions: three options are good actions, one is not.

## Mock 8 Q43 Merger arbitrage gain in a stock-for-stock deal: convert the long shares, close the short

You **B** · Answer **B**

### VIGNETTE (RELEVANT PART, UNEDITED)

First, Pasco reviews Fund 2, an event-driven hedge fund. The fund has a merger arbitrage trade involving a potential merger between two companies, Acquirer Company (AC) and Target Company (TC). AC has offered to buy TC in a stock-for-stock deal. The offer ratio is one share of AC for three shares of TC. AC was trading at $39 per share prior to the merger announcement, and its price decreased to $35 per share after the announcement. TC was trading at $10 per share prior to the merger announcement, and its price increased to $11 per share after the announcement. After the merger announcement, the hedge fund bought 30,000 shares of TC at $11 per share and sold 10,000 shares of AC at $35 per share.

**Question 43. If the merger between AC and TC closes successfully, Fund 2's merger arbitrage trade will result in a gain closest to:**

### OPTIONS

| | | | |
|---|---|---|---|
| **A** | $7,000. | | $7,000 is a one-sided payoff ($35 ÷ 3 − $11) × 10,000, which ignores that the conversion covers the short. |
| **B** | $20,000. | ✓ Correct | Correct: 10,000 × $35 − 30,000 × $11 = $20,000. |
| **C** | $90,000. | | $90,000 uses pre-announcement prices, not the trade prices. |

**Trigger** “The offer ratio is one share of AC for three shares of TC”
“bought 30,000 shares of TC at $11 per share and sold 10,000 shares of AC at $35 per share”

**Rule next time: 30,000 TC shares become 30,000 ÷ 3 = 10,000 AC shares, exactly the short position. Gain = short proceeds − cost of the long shares.**

### FORMULA / RULE

Merger arbitrage (stock deal): long target, short acquirer in the exchange ratio
Gain if the deal closes = short proceeds − cost of the target shares

### WHY THE ANSWER IS B

1. Long 30,000 TC at $11 = $330,000 paid.
2. Convert into 10,000 AC shares and deliver them against the short sold at $35 = $350,000.
3. Gain $20,000: **B**.

**Apply to other questions:** Right in this mock; keep the conversion step: shares of target ÷ exchange ratio.

## Mock 8 Q44 Main risk of merger arbitrage: the deal fails

You **B** · Answer **B**

### VIGNETTE (RELEVANT PART, UNEDITED)

First, Pasco reviews Fund 2, an event-driven hedge fund. The fund has a merger arbitrage trade involving a potential merger between two companies, Acquirer Company (AC) and Target Company (TC). AC has offered to buy TC in a stock-for-stock deal. The offer ratio is one share of AC for three shares of TC. AC was trading at $39 per share prior to the merger announcement, and its price decreased to $35 per share after the announcement. TC was trading at $10 per share prior to the merger announcement, and its price increased to $11 per share after the announcement. After the merger announcement, the hedge fund bought 30,000 shares of TC at $11 per share and sold 10,000 shares of AC at $35 per share.

**Question 44. The main risk of Fund 2's merger arbitrage trade is that:**

### OPTIONS

| | | | |
|---|---|---|---|
| **A** | the AC share price rises after the merger closes, which hurts the short AC position. | | After closing, the converted shares offset the short, so a higher AC price does not hurt a closed trade. |
| **B** | the merger is cancelled and the TC shares fall back toward their pre-announcement price. | ✓ Correct | Correct: deal cancelled, TC falls back and the short hedge no longer offsets. |
| **C** | the TC shares are converted into AC shares at the announced ratio, which locks in a loss. | | Conversion at the announced ratio is the expected path to a gain. |

**Trigger** “merger arbitrage trade”
“stock-for-stock deal”

**Rule next time: The trade is long target, short acquirer. It earns the spread if the deal closes, so the main risk is that the deal is cancelled.**

### FORMULA / RULE

> Merger arbitrage return = spread captured; risk = deal break

### WHY THE ANSWER IS B

1. Closing: shares convert and cover the short: gain.
2. Break: TC drops back to pre-deal price: loss.
3. **B**.

**Apply to other questions:** Right in this mock; keep: event-driven risk = event does not happen.